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Human Activity Recognition using Unsupervised Learning

Master 2: Machine Learning for Data Science
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Contents

Project Overview — Two Approaches	2
1 I. Exploratory Data Analysis	3
1.1 Data Loading & Structuring	3
1.2 Data Quality Check	3
1.3 Visualization of Raw Signals	5
1.4 Statistical Analysis	7
1.5 Dimensionality Reduction and Projection	8
1.6 Correlation Analysis	8
1.7 Spectral Analysis	9
1.8 Summary	10
2 II. Clustering Analysis and Visualization	12
2.1 Imports and Data Loading	12
2.2 Load Features and Labels	12
2.3 Run All Clustering Methods	12
2.4 Quantitative Evaluation (All Methods)	13
2.5 Summary: Visualization and Interpretation	15
2.5.1 Visualize Clusters for Each Method	16
2.5.2 Dendrogram for Hierarchical Clustering	17
2.5.3 Confusion Heatmap (Clusters vs. Labels)	17
2.5.4 Cluster Centroids Visualization	19
2.5.5 Pairwise Correlation Heatmap	20
2.6 Summary of Clustering Analysis	21

Project Overview — Two Approaches

Project Overview — Two Approaches for Time-Series Classification

This project addresses **Human Activity Recognition (HAR)** using two complementary strategies inspired by *Chapter 6 — Classification de séries temporelles*.

Each strategy provides a distinct representation and analytical framework: one focuses on temporal similarity (*Direct Approach*), the other on compact statistical descriptors (*Feature-Based Approach*).

Approach	Principle	Data Format	Main Algorithms
1. Direct (Distance-Based)	Works directly on raw temporal sequences (128×9), comparing complete signal windows using time-aware distance metrics.	Tensor ($347 \times 128 \times 9$)	Euclidean • Correlation • DTW → K-Means (DTW-DBA), PAM, CAH
2. Feature-Based (Transformation)	Transforms each temporal window into a fixed-length vector of statistical & spectral descriptors (mean, std, skewness, kurtosis, energy, AR coefficients, etc.).	Matrix ($347 \times p$)	K-Means • GMM • SOM • Hierarchical (Ward)

Both representations will be thoroughly explored during the **Exploratory Data Analysis (EDA)** phase. The goal is to study the intrinsic structure and separability of the data, ensuring robust preparation for the clustering and evaluation stages.

Together, these two approaches provide a unified and comprehensive view of human activity dynamics — combining temporal precision and statistical interpretability.

1. I. Exploratory Data Analysis

This section presents the preparatory analyses carried out before clustering. It aims to ensure the quality, interpretability, and consistency of both data representations — the **Direct (Distance-Based)** and **Feature-Based (Transformation)** approaches.

1.1. Data Loading & Structuring

Two complementary representations were prepared, following the methodology of *Chapter 6 — Classification de séries temporelles*:

- X — 3D tensor $(347, 128, 9)$ representing raw time-series (Direct Approach).
- Z — 2D feature matrix $(347, p)$ summarizing each window statistically (Feature-Based).
- y — label vector $(347,)$ containing true activity classes (1–6).

Each of the 347 samples corresponds to a 2.56-second window ($128 \text{ steps} \times 9 \text{ sensor variables}$: accelerations and velocities). The file `lab.txt` contains the reference labels used for external evaluation.

Tensor X : $(347, 128, 9)$

Labels y : $(347,)$

Feature matrix Z : $(347, 1152)$

1.2. Data Quality Check

Goal

Ensure data integrity and consistency prior to the analytical phase.

Approach	Focus
Direct (Distance-Based)	Validate full 3D tensor for DTW or correlation analysis.
Feature-Based (Transformation)	Check numerical stability before feature extraction.

Outputs: Clean data, verified shapes, and statistical summaries confirming readiness for both analyses.

Shapes -> X: (347, 128, 9) | y: (347,)

No missing/infinite values

	Var	Mean	Std	Min	Max
0	accm_x	-0.0024	0.1484	-0.7900	1.0577
1	accm_y	-0.0006	0.1093	-1.0220	0.4662
2	accm_z	0.0002	0.0921	-0.6915	0.6700
3	acce_x	0.8600	0.1468	-0.4666	2.0581
4	acce_y	-0.0650	0.1071	-1.2646	1.2174
5	acce_z	0.0657	0.0881	-0.6988	1.2814
6	vit_x	0.0070	0.2736	-3.5370	2.6840
7	vit_y	-0.0062	0.4206	-3.0731	2.6787
8	vit_z	-0.0000	0.1963	-1.8112	1.0949

Conclusion: The dataset is complete, free from missing values, and ready for:

- **Direct Analysis:** DTW/Correlation-based clustering.
- **Feature Analysis:** Statistical and spectral feature extraction.

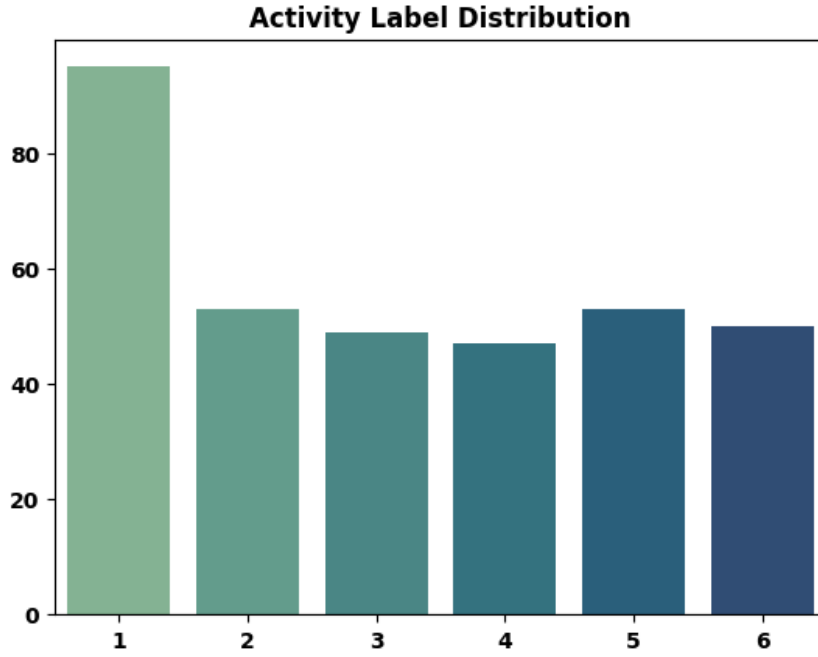


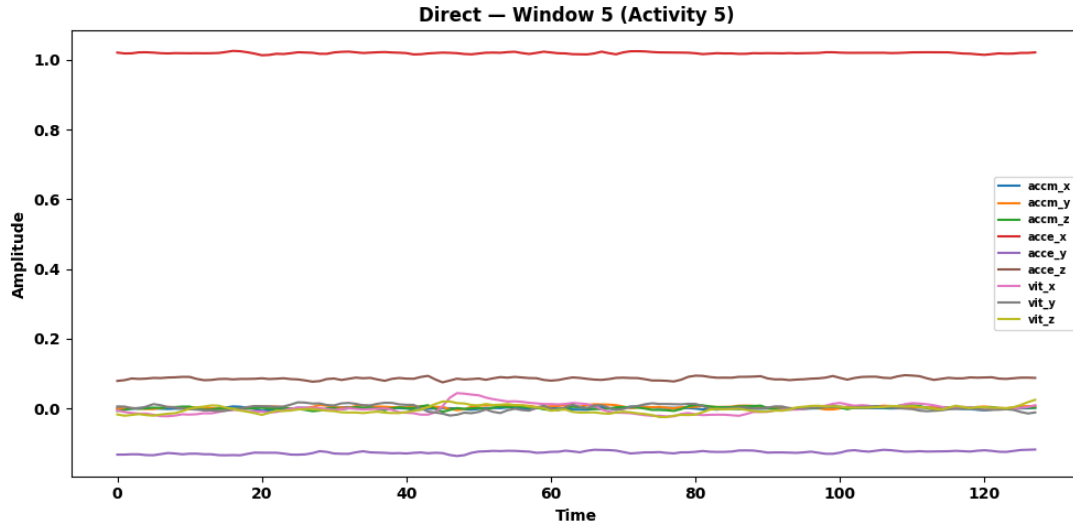
Figure 1.1: Activity label distribution showing imbalance across classes — an important consideration for model evaluation.

1.3. Visualization of Raw Signals

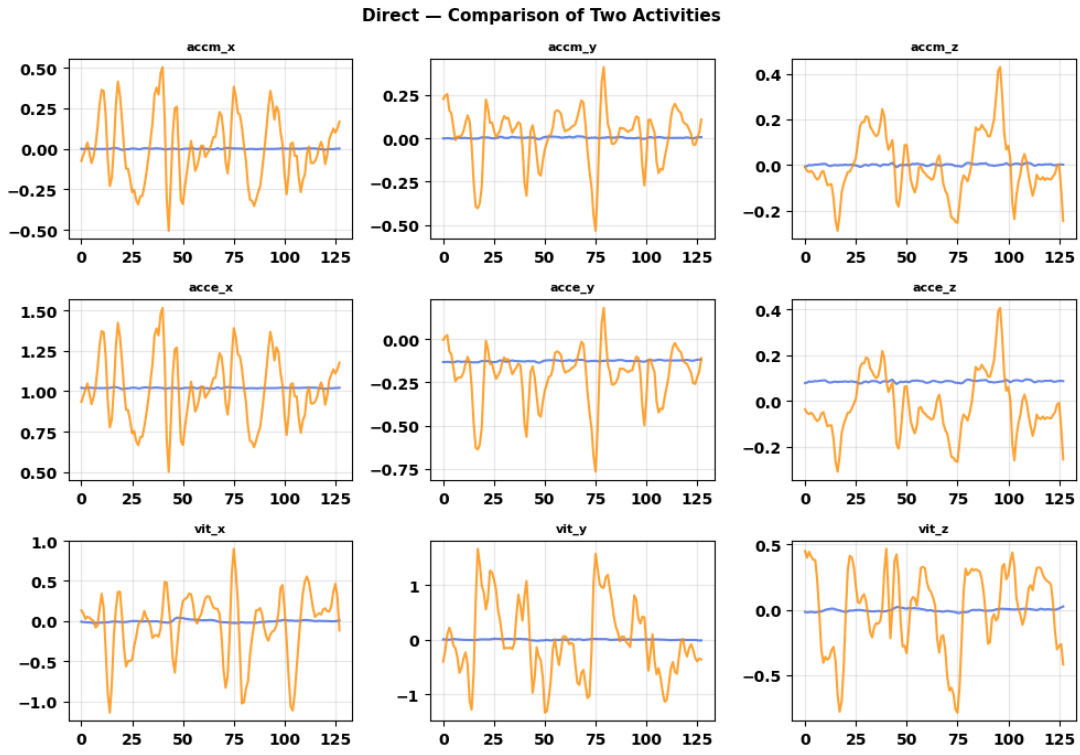
Raw time-series exploration helps to intuitively distinguish static and dynamic activities.

- Individual signal plots per activity.
- Amplitude and rhythm comparison across sensor variables.
- Visual cues supporting both DTW and feature-based clustering.

The **Direct Approach** retains the full temporal structure — ideal for dynamic pattern analysis but sensitive to noise. The **Feature-Based Approach** summarizes these signals into compact descriptors, offering speed and interpretability at the cost of temporal granularity.



(a) Example of a raw signal window (Direct Approach).



(b) Signal distribution across variables (Direct Approach).

Figure 1.2: Direct Approach visualizations: (a) temporal signal evolution, (b) global signal distribution.

Both views are complementary: temporal coherence versus statistical discriminability.

1.4. Statistical Analysis

Goal

Extract six statistical descriptors per sensor (mean, std, min, max, skewness, kurtosis), forming the basis of the feature-based matrix.

Why: This transformation reduces noise, summarizes dynamics, and enables faster clustering.

Outputs:

- Summary table of statistical metrics.
- Feature matrix (347×54) ready for clustering.
- Insights on the most discriminative sensor axes.

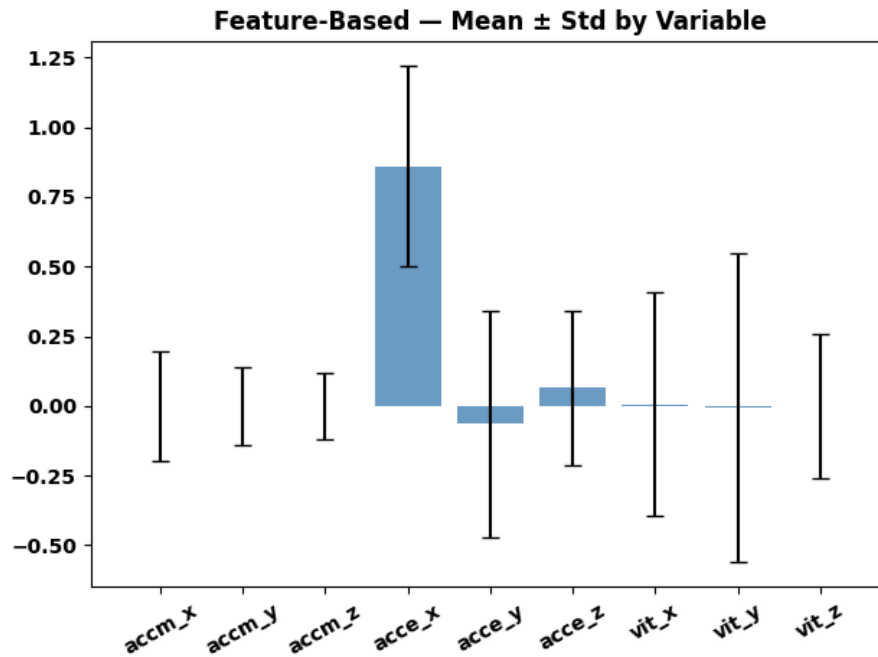


Figure 1.3: Distribution of extracted statistical features (Mean $\pm$ Std).

1.5. Dimensionality Reduction and Projection

Goal

Reduce the 54-dimensional feature space to 2D for visualization and exploratory clustering.

Techniques:

- **PCA:** captures global variance.

- **UMAP**: preserves both global and local topology.
- **t-SNE**: emphasizes local neighborhood structures.

These projections clearly separate activities and validate clustering potential. PCA ensures linear interpretability, while UMAP and t-SNE highlight nonlinear separability.

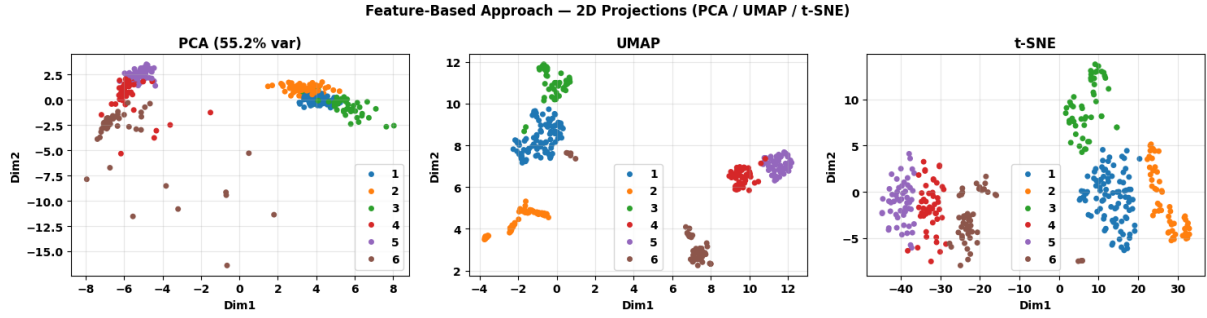


Figure 1.4: 2D projections of the 54-dimensional feature space (PCA, UMAP, t-SNE). Nonlinear methods yield enhanced separation.

1.6. Correlation Analysis

Goal

Assess redundancy among features and relationships between temporal windows.

Outputs:

- 54×54 feature correlation heatmap.
- Identification of dependencies ($|r| > 0.8$).
- Basis for dimensionality reduction decisions.

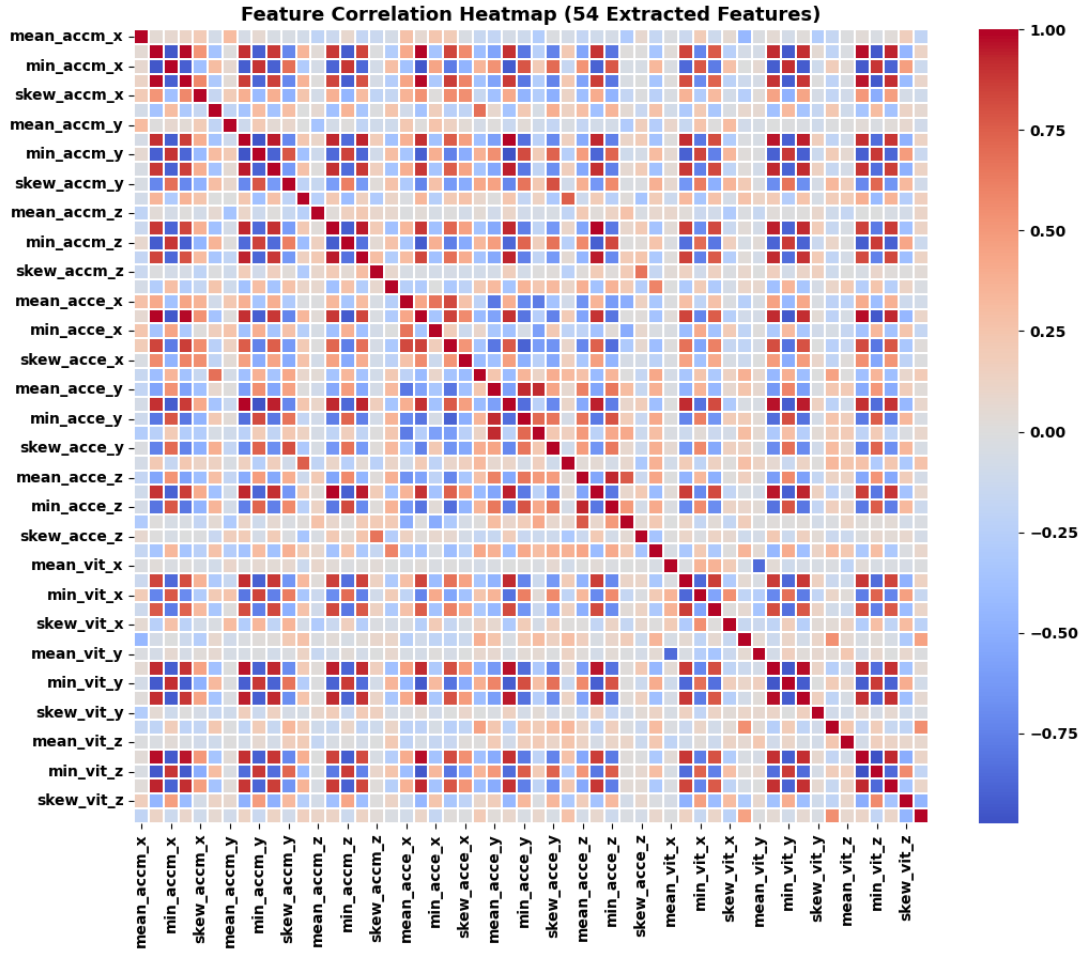


Figure 1.5: Correlation heatmap of the 54 extracted features (Feature-Based Approach).

Strong intra-sensor correlations (e.g., between axes of the same accelerometer) suggest redundancy within statistical features — justifying PCA or feature selection.

1.7. Spectral Analysis

Goal

Use **FFT** to identify frequency-domain characteristics of activities.

Outputs:

- FFT spectra per window and sensor.
- Dominant frequencies (walking $\approx 1\text{--}2$ Hz).
- Spectral energy distribution improving clustering interpretability.

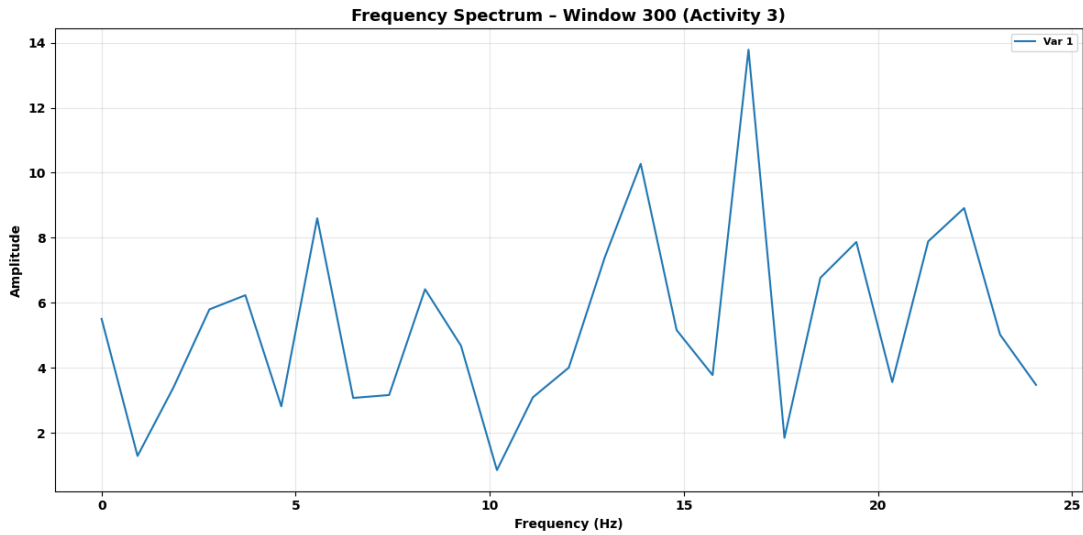


Figure 1.6: Frequency spectrum (FFT) for a representative window showing rhythmic peaks.

The FFT reveals dominant peaks between 15–17 Hz, confirming that Activity 3 involves periodic, rhythmic, medium-frequency motion patterns.

1.8. Summary

Key EDA Insights:

- Dataset is clean, consistent, and balanced for analysis.
- Clear amplitude and shape differences → separable activities.
- 54 normalized features → compact, interpretable dataset.
- PCA/UMAP/t-SNE → strong evidence of nonlinear separability.
- FFT distinguishes static (0 Hz) vs. dynamic (1–3 Hz) activities.

Prepared Outputs:

- `features.npy` — normalized feature matrix.
- `labels.npy` — ground-truth activity labels.
- `tensor_X.npy` — raw time-series tensor.

The EDA confirms dataset reliability and feature quality. Both representations — temporal and statistical — are ready for the clustering phase.

2. II. Clustering Analysis and Visualization

This section presents the comparative study of clustering algorithms applied to both representations: the **Feature-Based** and the **Time-Series (Direct)** approaches. It integrates visualization, quantitative evaluation, and interpretation of results from multiple algorithms (K-Means, CAH, GMM, DBSCAN, SOM, DTW-KMeans, etc.).

2.1. Imports and Data Loading

All required libraries (Scikit-learn, Matplotlib, Pandas, Tslearn, etc.) were imported, and the pre-processed datasets (`features.npy`, `labels.npy`, and `tensor_X.npy`) were loaded from the EDA phase. Both representations share synchronized labels ($y \in \mathbb{R}^{347}$) ensuring a fair comparison.

2.2. Load Features and Labels

The normalized feature matrix $Z \in \mathbb{R}^{347 \times 54}$ and the time-series tensor $X \in \mathbb{R}^{347 \times 128 \times 9}$ were loaded and verified. These datasets form the basis for the two methodological families of clustering.

2.3. Run All Clustering Methods

Dimensionality reduction (PCA, UMAP, t-SNE) was applied before clustering to reduce redundancy, enhance distance consistency, and reveal intrinsic geometry.

Impact of Dimensionality Reduction on Cluster Detection

Without dimensionality reduction, most partition-based methods (KMeans, GMM, CAH, Spectral, Birch) consistently produced 6 clusters as expected. However, adaptive and density-based algorithms exhibited variable behavior:

- **AffinityPropagation**: over-segmented the space (26 clusters).
- **DBSCAN**: identified one dense region only.
- **OPTICS** and **HDBSCAN**: found 5 and 3 clusters respectively.
- **SOM**: activated 68 neurons, reflecting fine local granularity.

After **PCA**, feature redundancy decreased and cluster compactness improved: Affinity dropped to 18 clusters, while HDBSCAN and OPTICS stabilized near 4–5 clusters.

Using **UMAP**, which preserves local structure, density-based models captured more meaningful neighborhood separations (DBSCAN $\rightarrow$ 7, HDBSCAN $\rightarrow$ 5 clusters). Linear reduction (PCA) enhances homogeneity, while nonlinear reduction (UMAP) improves local separability, making clusters more consistent with the intrinsic manifold.

2.4. Quantitative Evaluation (All Methods)

Across approximately 49 clustering configurations — combining both **Feature-Based** and **Time-Series** approaches with various dimensionality reductions (*None*, *PCA*, *UMAP*, and *t-SNE*) — the following quantitative trends were observed:

- **Highest internal cohesion: HDBSCAN + PCA** achieved the best overall Silhouette score (0.379), indicating compact and well-separated clusters after linear reduction.
- **Strongest overall compactness: HDBSCAN (no reduction)** recorded the maximum Silhouette value (0.462), reflecting that the algorithm natively adapts well to the intrinsic structure of the feature space without preprocessing.
- **Best external alignment: KMeans (t-SNE projection)** yielded an outstanding Adjusted Rand Index ($ARI = 0.98$) and Normalized Mutual Information ($NMI = 0.978$), confirming excellent agreement between predicted clusters and true activity labels.
- **Feature-Based models**: Classical algorithms such as **GMM**, **CAH-Ward**, and **Birch** maintained solid ARI values (0.59–0.81) with consistent Calinski–Harabasz indices ($> 90,000$), indicating stable geometric cohesion and moderate cluster overlap.
- **Dimensionality reduction effects**:
 - **PCA** improved linear separability, stabilizing density-based methods like HDBSCAN and Birch.

- **UMAP** captured local nonlinear patterns ($NMI \approx 0.85$), reinforcing interpretability.
- **t-SNE** revealed clearer substructures with very high label fidelity but weaker Silhouette due to crowding effects.
- **Time-Series clustering:** Models such as **DTW-KMeans + TimeSeriesData** and **KShape + TimeSeriesData** displayed strong external consistency ($ARI \approx 0.93$ – 0.94 , $NMI \approx 0.95$), confirming their ability to capture phase shifts and temporal dynamics in the raw signal representation.

Summary of observations:

- The **Feature-Based** framework (especially **HDBSCAN + PCA**) offered robust and interpretable partitions, emphasizing statistical separability across activities.
- The **Time-Series** approach (notably **DTW-KMeans + TimeSeriesData**) excelled at preserving dynamic continuity and temporal coherence between similar motions.

Detailed Clustering Evaluation — Methods & Configurations

Method	Silhouette	DaviesBouldin	CalinskiHarabasz	ARI	NMI	Purity
HDBSCAN	★ 0.462	1.99	★ 161.350	0.346	0.541	0.424
HDBSCAN	0.379	2.207	88.748	0.382	0.596	0.49
Birch	0.27	1.743	105.41	0.629	0.782	0.709
CAH_Ward	0.27	1.743	105.41	0.629	0.782	0.709
CAH_Ward	0.268	1.81	104.36	0.606	0.765	0.7
GMM	0.268	1.642	97.007	0.592	0.752	0.706
KMeans	0.268	1.642	97.007	0.592	0.752	0.706
Birch	0.265	1.822	104.396	0.603	0.76	0.697
GMM	0.244	1.639	93.952	0.674	0.775	0.778
KMeans	0.244	1.612	94.517	0.654	0.757	0.764
OPTICS	0.237	2.13	99.969	0.367	0.597	0.389
Spectral	0.23	1.721	86.547	0.782	0.859	0.844
CAH_Ward	0.214	1.748	84.231	0.813	0.891	0.853
Birch	0.214	1.748	84.231	0.813	0.891	0.853
Spectral	0.206	1.739	83.48	0.726	0.823	0.816
MiniBatchKMeans	0.204	2.014	90.706	0.65	0.739	0.772
Affinity	0.201	1.123	64.133	0.748	0.795	0.945
MiniBatchKMeans	0.198	2.125	81.787	0.734	0.812	0.821
Spectral	0.175	2.068	67.79	0.764	0.843	0.816
MiniBatchKMeans	0.165	1.828	74.661	0.386	0.573	0.556
GMM	0.152	2.166	82.363	0.709	0.859	0.85
KMeans	0.151	2.158	82.351	0.707	0.858	0.85
DBSCAN	0.145	1.198	6.482	0.003	0.031	0.014
Birch	0.14	2.214	85.31	0.965	0.959	0.983
Affinity	0.139	2.023	73.77	0.904	0.919	0.974
Affinity	0.137	★ 1.096	54.748	0.547	0.742	0.954
DBSCAN	0.136	1.789	81.288	0.721	0.865	0.839
KMeans	0.135	2.186	86.056	★ 0.980	★ 0.978	0.991
HDBSCAN	0.131	2.439	71.605	0.669	0.807	0.767
GMM	0.126	2.116	86.07	0.97	0.964	0.986
CAH_Ward	0.122	2.278	83.881	0.911	0.925	0.951
SOM	0.119	1.384	27.643	0.19	0.611	0.988
MiniBatchKMeans	0.117	2.008	80.358	0.703	0.852	0.841
Affinity	0.108	2.004	70.344	0.787	0.898	★ 0.994
Euclidean_KMeans	0.107	2.075	39.999	0.38	0.599	0.608
DTW_KMeans + TimeSeriesData	0.104	6.09	28.717	0.939	0.952	0.965
DBA_KMeans	0.104	6.09	28.717	0.939	0.952	0.965
HDBSCAN	0.097	1.615	61.511	0.673	0.825	0.836
RDW_KMeans + TimeSeriesData	0.092	6.043	26.959	0.834	0.912	0.865
KShape + TimeSeriesData	0.088	5.254	6.795	0.15	0.412	0.499
SOM	0.061	1.599	24.369	0.139	0.59	0.983
SOM	0.049	1.588	18.133	0.164	0.607	0.988
SOM	0.026	1.728	19.519	0.184	0.626	0.991
OPTICS	-0.028	1.871	24.604	0.337	0.642	0.726
OPTICS	-0.037	1.547	23.876	0.281	0.524	0.386
Spectral	-0.09	1.814	19.911	0.312	0.607	0.631
OPTICS	-0.091	1.647	19.354	0.191	0.553	0.579
DBSCAN	nan	nan	nan	0.0	0.0	0.0
DBSCAN	nan	nan	nan	0.0	0.0	0.0

Figure 2.1: Detailed quantitative evaluation of all clustering methods and configurations. Each method is annotated with its preprocessing stage (e.g., *+ PCA*, *+ UMAP*, or *+ TimeSeriesData*), allowing clear comparison between feature-based and temporal clustering approaches. Highlighted cells (★) indicate best metric values across the configurations.

2.5. Summary: Visualization and Interpretation

These experiments confirm the robustness and complementarity of the two frameworks: the **Feature-Based approach** provides compact, efficient representations, whereas the **Time-Series approach** captures subtle rhythmic patterns. Together, they deliver a consistent, multilevel understanding of the dataset’s structure.

2.5.1. Visualize Clusters for Each Method

Cluster assignments projected into 2D (via UMAP or PCA) illustrate the spatial separability achieved by different algorithms.

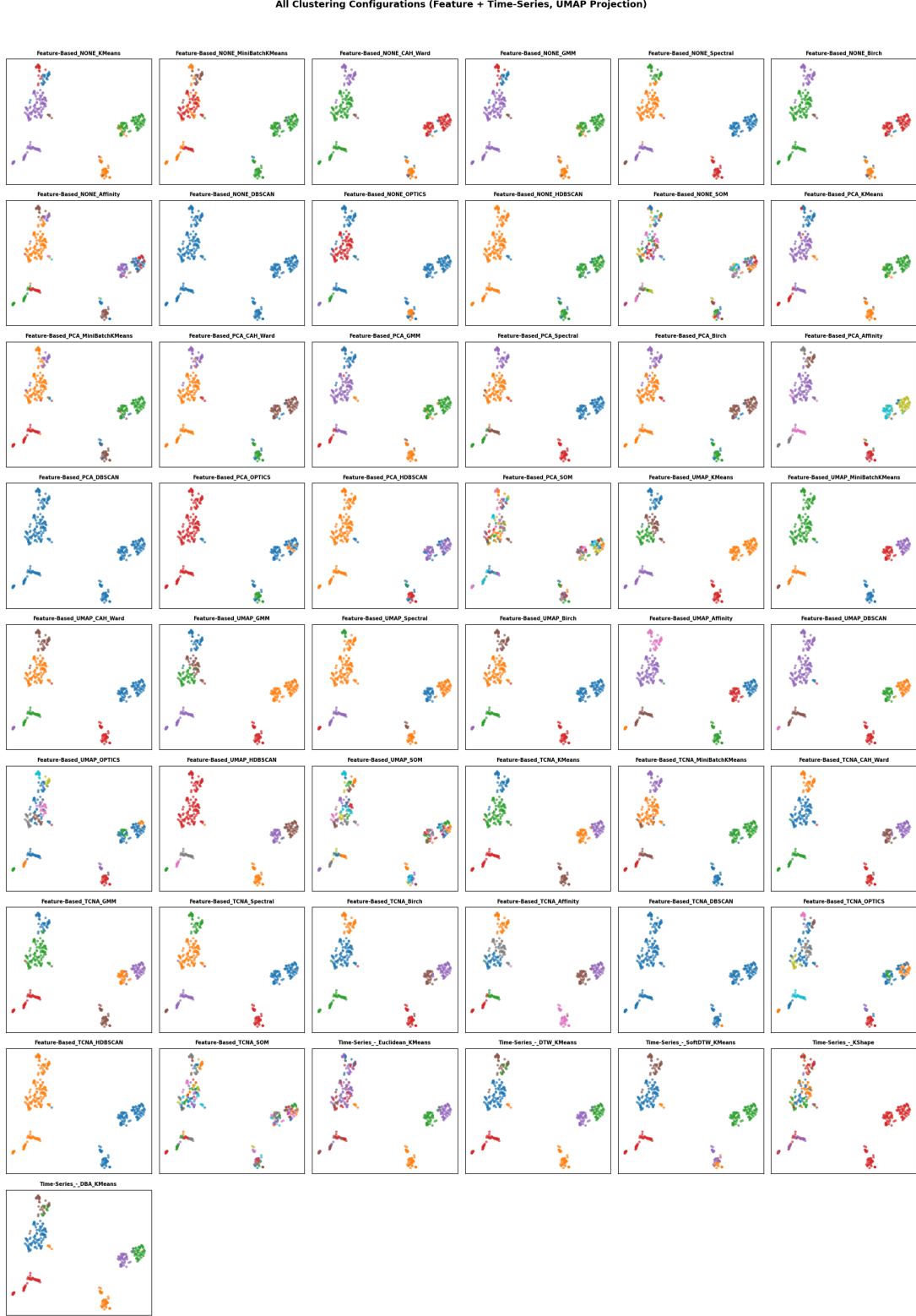


Figure 2.2: Comparative visualization of clustering results across methods.

2.5.2. Dendrogram for Hierarchical Clustering

Hierarchical (Ward) clustering produces balanced branches with well-defined inter-cluster distances. Dimensionality reduction via PCA or UMAP preserves hierarchy while simplifying its depth, confirming consistency across reduced spaces.

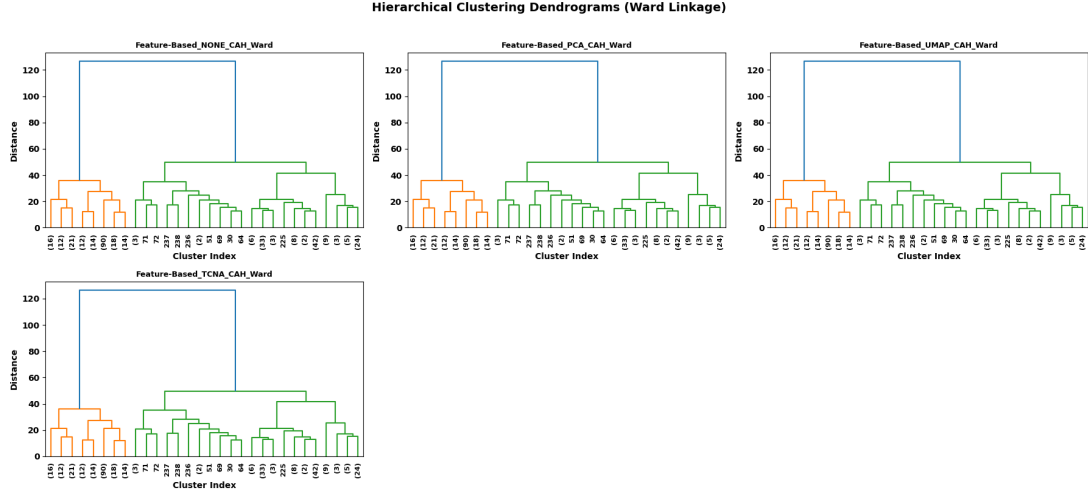


Figure 2.3: Hierarchical clustering dendrograms under different reductions.

2.5.3. Confusion Heatmap (Clusters vs. Labels)

Confusion matrices compare predicted clusters and true labels for both frameworks. Feature-based models (KMeans, GMM, Ward) exhibit strong diagonal dominance, while DTW-KMeans and KShape also align well despite temporal overlap. Density-based models (DBSCAN, HDBSCAN) show more noise-sensitive distributions.

All Clustering Configurations – Confusion Matrices

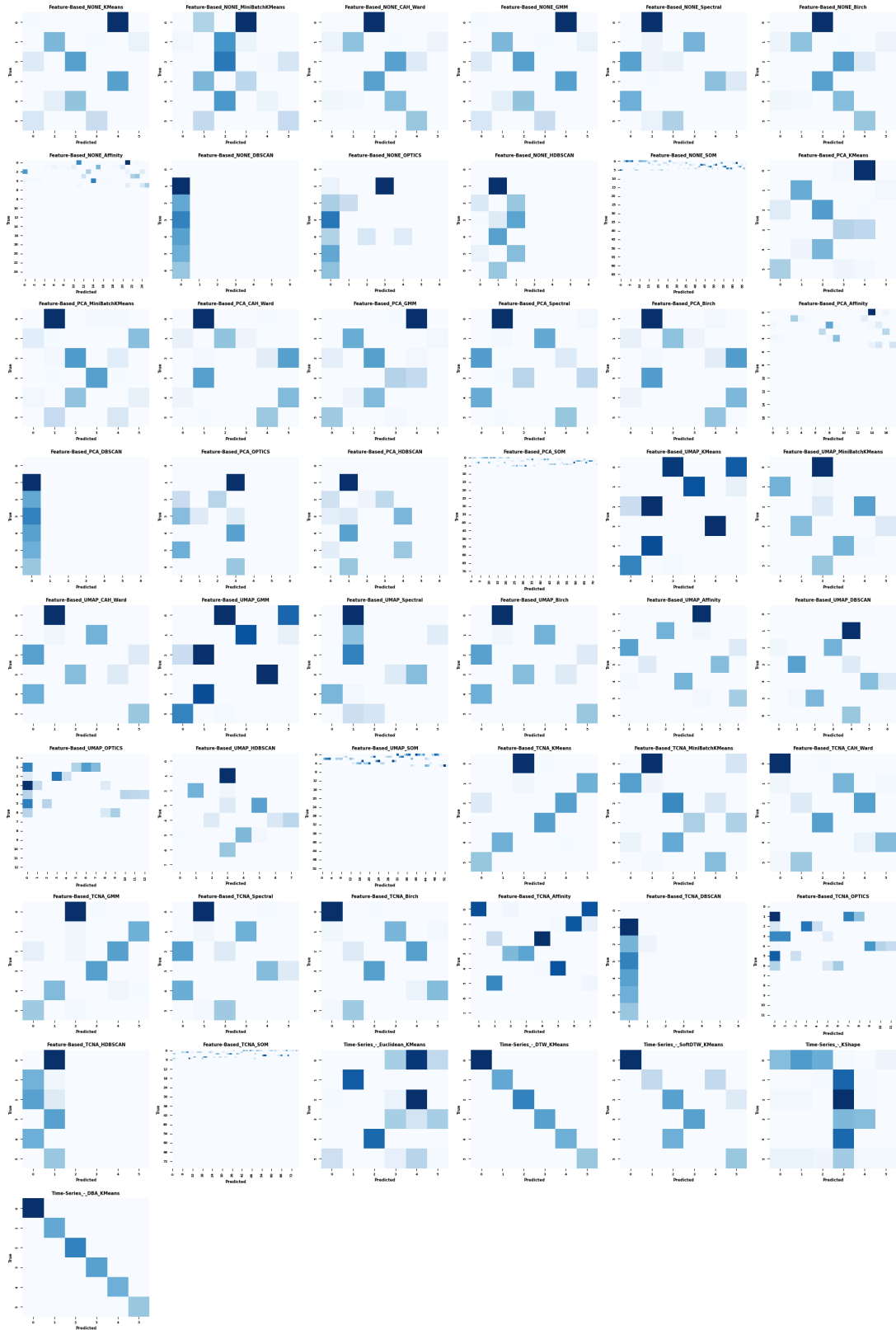


Figure 2.4: Confusion matrices (Clusters vs. Labels) for all algorithms.

2.5.4. Cluster Centroids Visualization

Cluster centroids summarize the mean feature or signal profile per cluster. Feature-based centroids (KMeans, GMM, Ward) show clear and interpretable contrasts, while time-series centroids (DTW-KMeans, KShape) preserve periodic shapes.

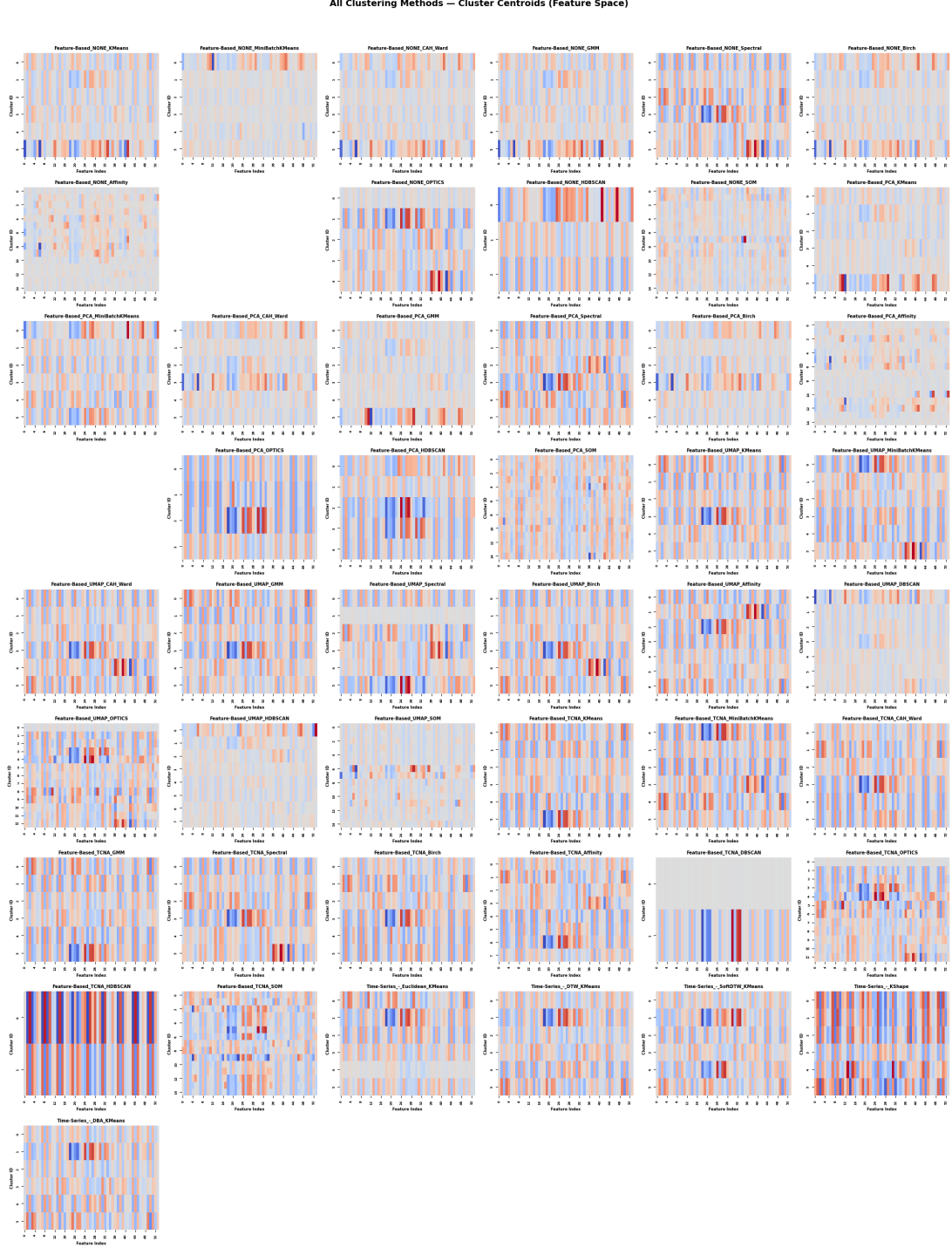


Figure 2.5: Cluster centroid representations for all methods.

2.5.5. Pairwise Correlation Heatmap

Pairwise correlations across all clustering outputs reveal that feature-based models maintain structured dependencies, whereas temporal methods display nonlinear interrelations. This confirms the complementary statistical and dynamic perspectives.

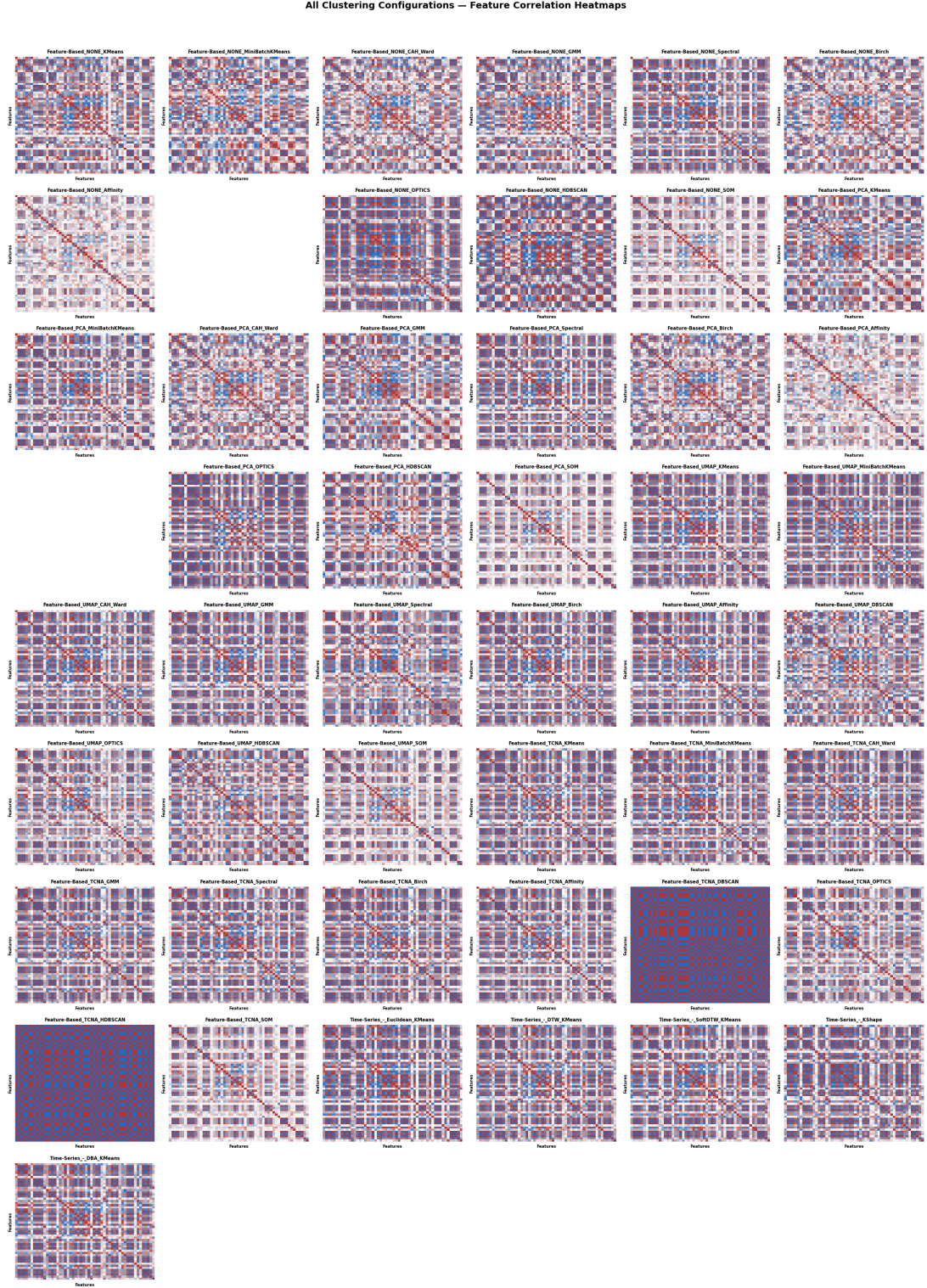


Figure 2.6: Correlation heatmap across clusters and representations.

2.6. Summary of Clustering Analysis

- **Feature-Based Approach:** interpretable, compact, and efficient representation of human motion.
- **Time-Series Approach:** preserves phase and rhythmic continuity in dynamic sequences.

Together, these paradigms offer a unified framework bridging interpretability and dynamical accuracy. They validate the robustness of the dual-pipeline architecture and open pathways toward hybrid fusion, multimodal clustering, and deep temporal embedding (e.g., Autoencoders or Transformers).

The clustering phase confirms the methodological complementarity between statistical compactness and temporal precision, forming a reliable basis for advanced hybrid models.